Sampling & Estimation

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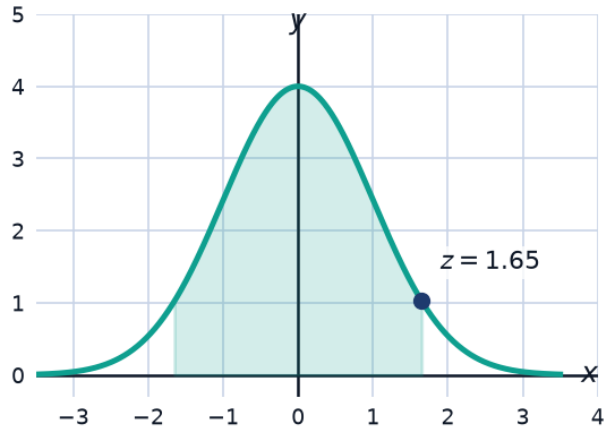
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the equation $\ln(3x+1) - 2\ln 2 = \ln(x-1)$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The masses, x grams, of 50 packets of tea gave $\sum x = 5090$ and $\sum x^2 = 518380$. (a) Calculate unbiased estimates of the population mean and variance. (b) Calculate a 95% confidence interval for the population mean.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A sample of size 100 from a population (known $\sigma = 20$) gives $\bar{x} = 142$. Find a 99% CI for μ and state what "99% confidence" means.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Sampling & Estimation · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

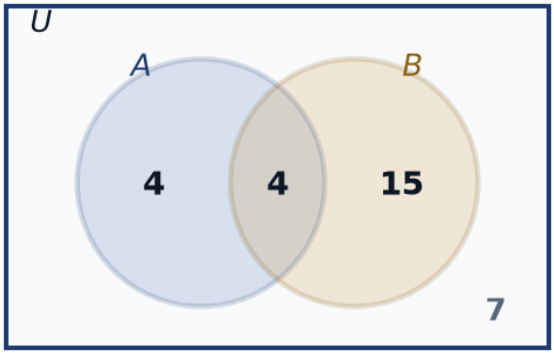
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Determine whether the lines $\mathbf{r}_1 = (1; 2; -1) + \lambda(2; 1; 3)$ and $\mathbf{r}_2 = (0; 5; 4) + \mu(-1; 2; 1)$ intersect or are skew.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the binomial expansion of $\frac{1}{\sqrt{4-x}}$ in ascending powers of x up to and including the term in x^2. State the set of values of x for which the expansion is valid.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Use the substitution $u = 2x + 1$ to find the exact value of $\int_0^4 \frac{x}{\sqrt{2x+1}}\,dx$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A random variable X has distribution $\text{Po}(\lambda)$. A single observation of X is taken to test $H_0 : \lambda = 6$ against $H_1 : \lambda < 6$ at the 5% level. (a) Find the critical region. (b) State the actual significance level. (c) Find the probability of a Type II error if the true value of λ is 3.

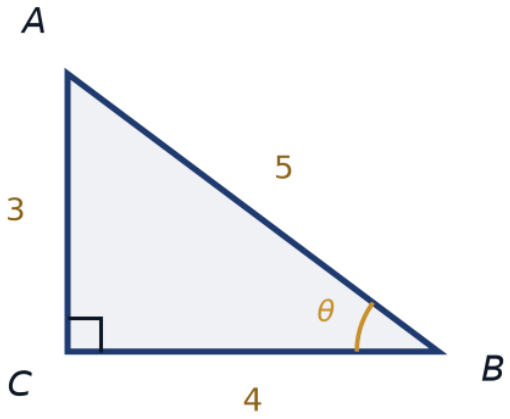
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the equation $2\cos 2x = 1 + \sin x$ for $0 \le x \le 2\pi$ radians, giving answers to 3 significant figures.

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10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The curve C has equation $x^3 - 2xy + y^2 = 4$. Find the coordinates of the two points on C where the tangent is parallel to the y-axis (vertical).



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The number of accidents on a highway follows a Poisson distribution with mean 1.6 per week. (a) Find the probability of at least 3 accidents in a 2-week period. (b) Given that at least one accident occurs in a week, find the probability that there are exactly 2 accidents in that week.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

On an Argand diagram, the point P represents the complex number z such that $|z - (2 + 3i)| \leq 2$. Find the greatest possible value of $|z|$ for points in this region.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A student calculates $\int_1^3 \frac{4}{2x+1}\,dx$ by rounding $\ln 7 \approx 1.95$ and $\ln 3 \approx 1.10$ at the start, obtaining $2(1.95 - 1.10) = 1.70$. State the exact value of the integral and calculate the absolute error introduced by premature rounding.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The equation $x^3 - x - 1 = 0$ has a root near $x = 1.3$. (a) Show that the iterative formula $x_{n+1} = (x_n + 1)^{1/3}$ converges by considering $g'(x)$. (b) Starting with $x_1 = 1.3$, calculate x_2 and x_3 to 4 decimal places.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

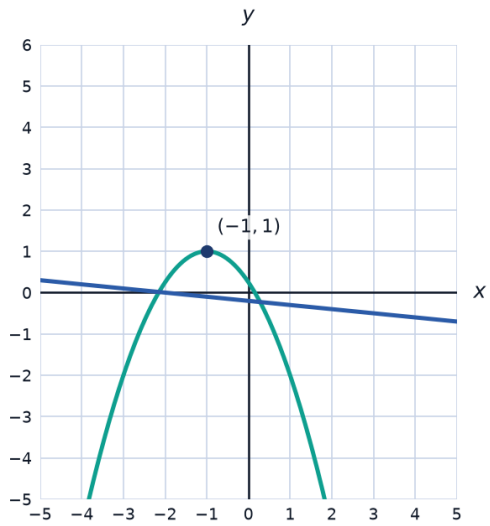
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 15.01 Hypothesis Testing

Variables x and y satisfy $y = ax^b$. A graph of $\ln y$ against $\ln x$ is a straight line passing through $(0, 1.20)$ and $(4, 6.80)$. Find the values of the constants a and b to 3 significant figures.

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Exponentials & Logarithms

$f(x) = cx^2$ for $0 \le x \le 3$, zero otherwise. Find c, $E(X)$, and $P(1 \le X \le 2)$.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Advanced Trigonometry

A manufacturer claims that the mean breaking strength of cables is at least 80 kN. A sample of 16 cables gives $\bar{x} = 77.5$ kN with population SD 5 kN. Test this claim at 1% significance.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Differentiation

Lines ℓ_1 and ℓ_2 have direction vectors $\mathbf{d_1} = (2; 1; -1)$ and $\mathbf{d_2} = (1; -1; 3)$. Find the acute angle between them.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Integration

Express $\frac{3x+1}{(x+1)(x-2)}$ in partial fractions.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Sequences & Series

Cars arrive at a toll booth at an average rate of 3 per minute. Find the probability that (a) exactly 5 arrive in one minute, (b) at least 2 arrive in 30 seconds.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the unbiased estimator of the population variance σ^2 calculated from a random sample of size n?

a. $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$

b. $s^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$

c. $s^2 = \frac{1}{n+1}\sum(x_i - \bar{x})^2$

d. $s^2 = \sum(x_i - \bar{x})^2$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A random sample of 10 observations has $\sum x = 80$ and $\sum x^2 = 724$. Calculate the unbiased estimate of the population variance.

a. 9.33

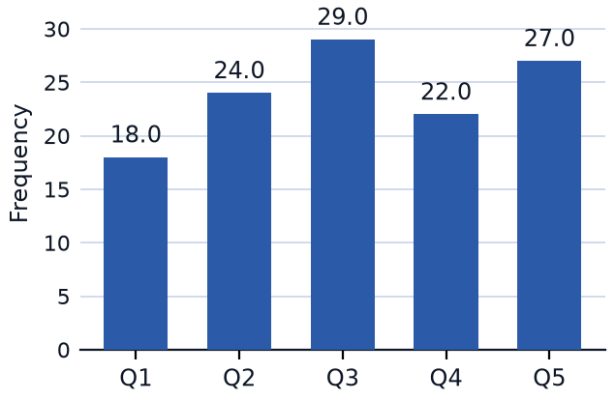
b. 8.40

c. 10.50

d. 7.20

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If the biased sample variance is $v = 18$ from a sample of size $n = 25$, find the unbiased estimate s^2.



a. 18.75

b. 17.28

c. 18.00

d. 19.50

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

State the Central Limit Theorem for the distribution of the sample mean $\bar{X}$.

a. For large n, $\bar{X}$ is approximately normal $N(\mu, \sigma^2/n)$ regardless of the population distribution

b. For any n, $\bar{X}$ is strictly normal

c. For large n, the population becomes normal

d. $\bar{X}$ has mean $n\mu$ and variance $n\sigma^2$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A population has mean 100 and standard deviation 15. A sample of size $n = 36$ is taken. What is the standard error of the sample mean?

a. 2.5

b. 15

c. 4.17

d. 0.417

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Using the distribution from the previous question ($\mu = 100$, SE $= 2.5$), find $P(\bar{X} > 105)$.

a. 0.0228

b. 0.0500

c. 0.0456

d. 0.0100

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A sample of size $n = 64$ from a normal population with known $\sigma = 8$ has sample mean $\bar{x} = 52.0$. Find the 95% confidence interval for μ.

a. $[50.04, 53.96]$

b. $[49.50, 54.50]$

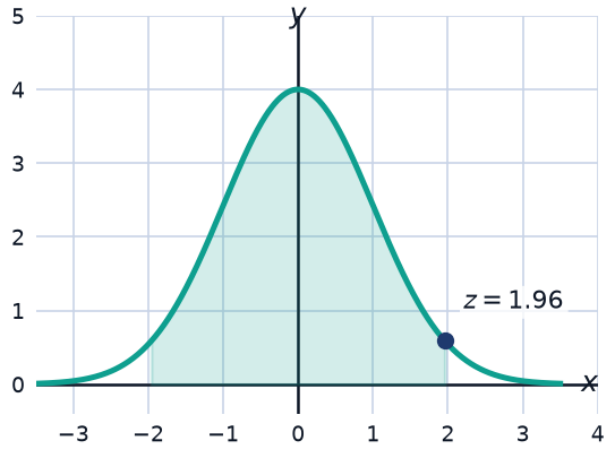
c. $[50.36, 53.64]$

d. $[51.00, 53.00]$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the critical value z used to construct a 99% confidence interval for the population mean?



a. 2.576

b. 1.960

c. 1.645

d. 2.326

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How large must a sample be so that the width of a 95% confidence interval for μ is at most 2 units, given $\sigma = 10$?

a. At least 385

b. At least 100

c. At least 196

d. At least 250

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

When σ^2 is unknown but the sample size $n = 100$ is large, what is substituted for σ^2 in calculating the confidence interval?

a. The unbiased sample variance estimate s^2

b. The sample range divided by 4

c. The median absolute deviation

d. A default value of 1.0

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A sample of 100 lightbulbs has mean life $\bar{x} = 1200$ hours and unbiased sample variance $s^2 = 2500$. Find the 90% confidence interval for the true mean life.

a. $[1191.8, 1208.2]$

b. $[1180.0, 1220.0]$

c. $[1190.2, 1209.8]$

d. $[1195.0, 1205.0]$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the correct probabilistic interpretation of a 95% confidence interval $[L, U]$ for μ?

a. In repeated sampling, 95% of such constructed intervals will contain the true parameter μ

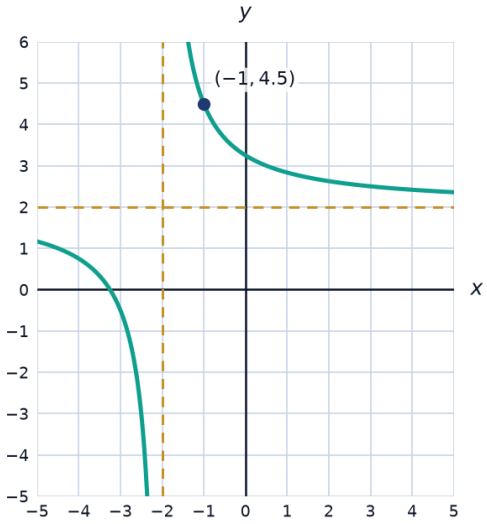
b. There is a 95% probability that μ lies in this specific interval

c. 95% of population values fall between L and U

d. The sample mean falls in $[L, U]$ 95% of the time

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given $\bar{x} = 61$, $z = 1$, $\sigma = 12$, $n = 4$, find the upper confidence limit $\bar{x} + E$.



a. 61

b. 67

c. 68

d. 55

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A required margin of error is $E = 4$, with $z = 1$, $\sigma = 8$. Find the required sample size $n = (z\sigma/E)^2$.

a. 8

b. 5

c. 2

d. 4

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A one-tail (upper) hypothesis test has critical value $z = 2.576$ and test statistic $z = 3$. What is the decision?

a. Accept H_1 unconditionally

b. Do not reject H_0

c. Reject H_0

d. Insufficient information

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Expand $\cos(x + 45^\circ)$ in terms of $\sin x$ and $\cos x$.

a. $\cos x \cdot \frac{\sqrt{2}}{2} + \sin x \cdot \frac{\sqrt{2}}{2}$

b. $\cos x \cdot \frac{\sqrt{2}}{2} - \sin x \cdot \frac{\sqrt{2}}{2}$

c. $\sin x \cdot \frac{\sqrt{2}}{2} - \cos x \cdot \frac{\sqrt{2}}{2}$

d. $\cos x \cdot \frac{\sqrt{2}}{2} - \sin x \cdot \frac{\sqrt{2}}{2} + 0$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Given $\mathbf{a} = (2, -6, -9)$ and $\mathbf{b} = (2, 3, 6)$, find $\cos\theta$, where θ is the angle between them.

a. $\frac{9}{77}$

b. $\frac{68}{77}$

c. $-\frac{34}{9}$

d. $-\frac{68}{77}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Differentiation

A curve is defined parametrically by $x = 2t^2$, $y = 3t^3$. Find $\frac{dy}{dx}$ at $t = 1$.

a. $\frac{7}{3}$

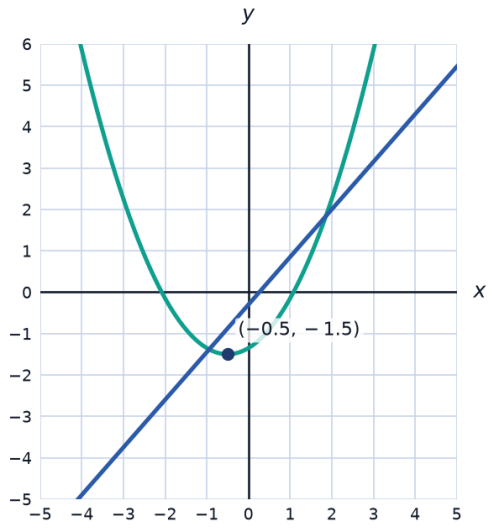
b. $\frac{9}{4}$

c. $-\frac{9}{4}$

d. $\frac{4}{9}$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Integration

Find the Cartesian equation of the locus $|z - 2| = |z - (-2)|$.



a. x=0

b. x=2

c. x=1

d. y=0

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Sequences & Series

Given $z = r(\cos 90^\circ + i\sin 90^\circ)$, find the principal argument (in degrees, $0 \le \theta < 360$) of z^3.

a. 270

b. 0

c. 274

d. 90

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Vectors

Solve $e^{2x} - 22e^x + 112 = 0$ for x. Give the larger solution in exact form.

a. $\ln 22$

b. $\ln 112$

c. $\ln 14$

d. $\ln 8$

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Given $\sin\theta = \frac{3}{5}$, with θ in Quadrant I, find $\tan 2\theta$.

a. $\frac{24}{7}$

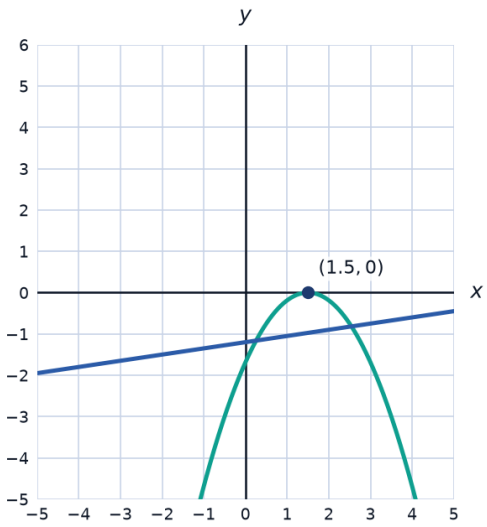
b. $\frac{7}{25}$

c. $-\frac{24}{7}$

d. $\frac{24}{25}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Complex Numbers

Solve $x^2 - 12x + 52 = 0$. Find the root with positive imaginary part.



a. 4+6i

b. -6+4i

c. 6+4i

d. 6-4i

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions: Modulus & Inverse

A sequence satisfies $u_{n+1} = \frac{1}{3}u_n + 1$ and converges to a limit L. Find L.

a. $\frac{5}{2}$

b. $-\frac{3}{2}$

c. 1

d. $\frac{3}{2}$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Numerical Methods

Given $\sin\theta = \frac{5}{13}$, with θ in Quadrant II, find $\sin 2\theta$.

a. $\frac{120}{169}$

b. $\frac{25}{169}$

c. $-\frac{120}{169}$

d. $\frac{119}{169}$

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